

Foundation Of Software Development :-

Assignment Solution:-

Q1 : Your friend wants to start a career in software but thinks only people from big cities or top colleges can get jobs in IT companies. Based on what you have learned about the software industry and career paths, what would you tell your friend? Explain what kind of roles are available and what skills matter most to get started.

Ans :- I would tell my friend that anyone can build a successful career in software, regardless of whether they come from a small town or a top college. Many IT companies hire candidates based on their skills, projects, and problem-solving ability rather than their location.

Some common software roles are:

- Software Developer
- Web Developer
- Mobile App Developer
- Software Tester (QA)
- UI/UX Designer
- DevOps Engineer
- Data Analyst

The most important skills are programming, logical thinking, communication, teamwork, and continuous learning. Building projects, earning certifications, and practicing coding regularly can help anyone get started in the software industry.

Q2 : A small shop owner in your town wants to create a mobile app so that customers can order products online. He asks you — "How will this app actually be built? Who all will work on it?" Explain to him the different roles involved in building software and how they work together as a team.

Ans :- To build a mobile app, different people work together as a team.

First, the shop owner explains what features the app should have. A business analyst understands these requirements and prepares a plan. A UI/UX designer creates the design so the app looks attractive and is easy to use. Frontend developers build the screens that customers use, while backend developers create the server and database that store customer and product information. Software testers check the app for mistakes or bugs before it is released. A project manager makes sure everyone completes their work on time.

When all these people work together, the app is developed successfully.

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Q3 : Your cousin is learning to code and wrote a program in Python. She clicked "Run" and got an error saying "SyntaxError". She is confused and asks you — "I wrote everything correctly, why is the computer not understanding me?" Explain to her why computers cannot directly understand what we write, and what happens between writing code and the computer actually running it.

Ans :- Computers cannot understand Python or any other programming language directly because they only understand machine language, which is made up of 0s and 1s.

When my cousin writes Python code and clicks **Run**, the Python interpreter first checks whether the code follows Python's rules. If there is a mistake, such as a missing bracket, quotation mark, or colon, it shows a **Syntax Error**. If the code is correct, the interpreter converts it into instructions that the computer can understand and then executes it to produce the output.

So, the computer is not confused—it simply needs code that follows the correct syntax.

Q4 :- You are asked to write step-by-step instructions (algorithm) for this problem: "A customer is buying items from an online store. If the total bill is more than ₹500, give a 10% discount. Otherwise, no discount." Write the solution first in plain English steps, and then convert it into simple pseudocode.

Ans:-

Algorithm:-

1. Start.
2. Enter the total bill amount.
3. Check if the bill is more than ₹500.
4. If yes, calculate a 10% discount.
5. Subtract the discount from the total bill.
6. If not, keep the bill amount the same.
7. Display the final amount.
8. Stop.

Pseudocode :-

START

INPUT bill

IF bill > 500 THEN

 discount = bill * 0.10

 finalBill = bill - discount

ELSE

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finalBill = bill

END IF

PRINT finalBill

STOP

Q5 :- Your younger brother opens his phone browser and types "www.flipkart.com". Within 2 seconds, the Flipkart homepage appears on his screen. He asks you — "How did this happen so fast? Where did this page come from?"

Ans :- When my brother types www.flipkart.com into the browser, the browser first asks the **DNS (Domain Name System)** to find the website's IP address. After getting the IP address, the browser connects to Flipkart's server using **HTTPS**, which provides a secure connection.

The browser then sends a request to the server asking for the webpage. The server responds by sending the website files, such as HTML, CSS, JavaScript, and images. Finally, the browser combines these files and displays the Flipkart homepage on the screen. This entire process happens within a few seconds because modern networks and servers are very fast.

Q6 :- : A startup founder tells you — "I want to build a food delivery app like Zomato. I don't want to buy expensive servers and computers. Is there any other way?" Explain to him what cloud computing is, why it is useful for startups, and give a simple example of how cloud services work.

Ans :- Cloud computing means using servers, storage, databases, and other computing services over the internet instead of buying and maintaining your own computers.

This is very useful for startups because they don't have to spend a lot of money on hardware. They only pay for the services they use, and they can easily increase or decrease resources as their business grows.

For example, if someone builds a food delivery app, they can host it on cloud platforms like AWS, Microsoft Azure, or Google Cloud. Whenever customers place orders, the cloud servers process the requests and store the data securely.

Q7:- What is the difference between a compiler and an interpreter? Explain in your own words with a simple real-life comparison .

Ans :- A compiler translates the entire program into machine language before running it, while an interpreter translates and executes the program one line at a time.

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A simple example is translating a book. A compiler is like translating the whole book before anyone reads it. An interpreter is like translating each sentence while someone is reading the book. Because of this, compiled programs usually run faster than interpreted programs.

Q8:- What is the Software Development Life Cycle (SDLC)? Name all its phases and write one line about what happens in each phase.

Ans :- The Software Development Life Cycle (SDLC) is a step-by-step process used to develop software in an organized way.

The phases are:

1. **Requirement Gathering** – Understand what the customer wants.
2. **Planning** – Decide the budget, timeline, and resources.
3. **Design** – Create the structure and design of the software.
4. **Development** – Write the program code.
5. **Testing** – Find and fix errors.
6. **Deployment** – Release the software to users.
7. **Maintenance** – Update the software and fix issues after release.

Q9 :- What is the difference between frontend development and backend development? Give one real example to show how both work together in a single application.

Ans:- Frontend development focuses on the part of an application that users can see and interact with. Backend development handles the server, database, and the logic that works behind the scenes.

For example, in an online shopping app, the frontend displays products, the shopping cart, and the payment page. The backend stores product information, checks customer details, processes payments, and updates the database. Both frontend and backend work together to make the application function properly.

Q10 :- What is the difference between HTTP and HTTPS? Why is HTTPS important when you are using a banking or payment website?

Ans :- HTTP and HTTPS are protocols used to transfer information between a web browser and a website. The main difference is that HTTP does not encrypt data, while HTTPS encrypts it using SSL/TLS, making the communication secure.

HTTPS is very important for banking and payment websites because it protects personal information such as passwords, bank account details, and card numbers from hackers. It

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also helps users know they are connected to the genuine website, making online transactions much safer.